DISTRIBUTED SYSTEMS

Presented By:
DR. Ahmed Al-Badawy
Lecturer, Computer Science Department

LECTURE OBJECTIVES

- Mutual Introduction
- Introduction To Distributed Systems
- Course Objectives and outcomes
- Examples
- Prerequisite courses
- Textbooks and Related references
- Communication Platform
- Grading Policies
- Summary

INTRODUCTION TO DISTRIBUTED SYSTEMS

A Distributed Systems course provides a comprehensive understanding of how modern software systems operate across multiple computers connected through a network. Instead of relying on a single centralized machine, distributed systems use multiple independent nodes that collaborate to achieve common goals. These nodes communicate, coordinate, and share resources to deliver scalable, reliable, and high-performance services. The course explores how complex applications such as cloud platforms, banking systems, social networks, healthcare platforms, and large-scale enterprise systems are designed to operate seamlessly across geographically distributed environments.

INTRODUCTION TO DISTRIBUTED SYSTEMS

The course introduces the fundamental principles behind distributing computation and data across multiple machines, including communication protocols, synchronization, coordination, fault tolerance, and consistency. It explains how systems handle failures gracefully, maintain data integrity, and ensure continuous availability even when individual components fail. Students learn how distributed systems achieve scalability by adding more machines, improve performance through parallel processing, and increase reliability through redundancy and replication.

COURSE OBJECTIVES AND OUTCOMES

- The course aims to:
  1. Understand the fundamental principles of distributed computing and system coordination
  2. Explain how independent computers communicate and collaborate over networks
  3. Learn techniques for scalability, fault tolerance, and high availability
  4. Study distributed data management, consistency, and synchronization
  5. Analyze real-world distributed architectures such as cloud platforms and microservices

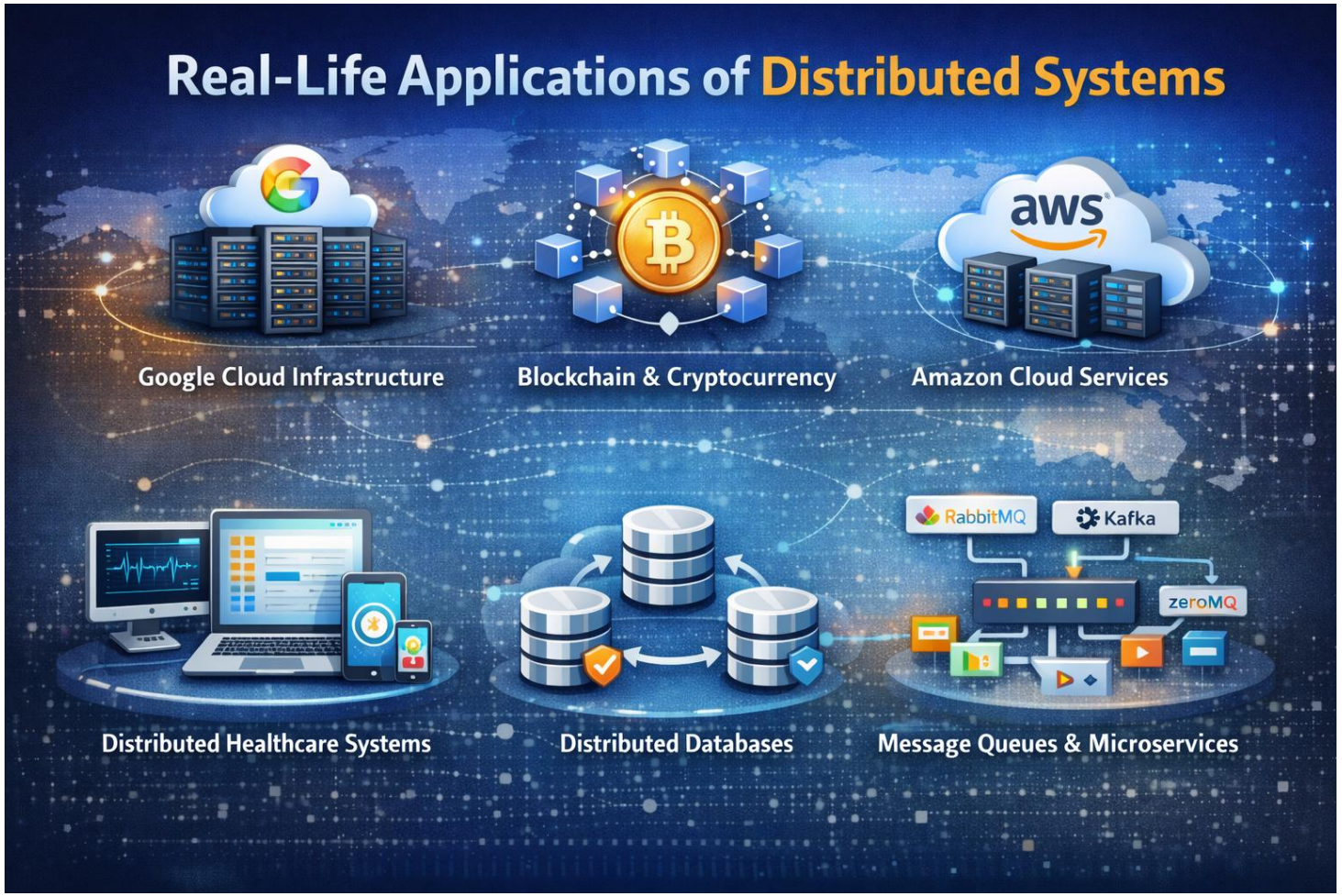
COURSE OBJECTIVES AND OUTCOMES

The outcomes for that course will be:

1. Explain the fundamental concepts and architectures of distributed systems, including client-server, peer-to-peer, and microservices models.
2. Design scalable and reliable distributed systems that ensure high availability and fault tolerance.
3. Implement communication and coordination between distributed components using APIs, remote procedure calls, and messaging systems.
4. Apply data management techniques, including replication, synchronization, and consistency models in distributed environments.
5. Analyze and evaluate distributed system performance, reliability, and scalability, and propose appropriate architectural solutions.

REAL LIFE DISTRIBUTED SYSTEMS APPLICATIONS

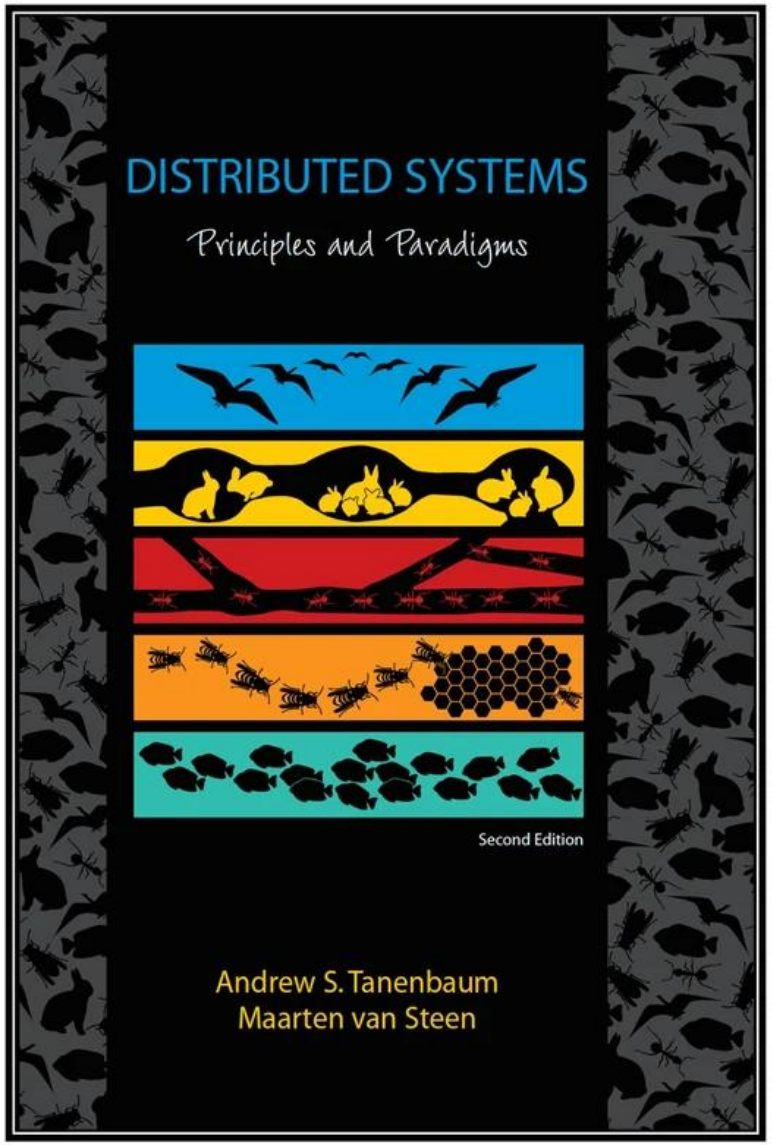
1. Distributed systems are used in:
2. Cloud platforms (AWS, Azure, Google Cloud)
3. Banking systems and financial transactions
4. Microservices-based applications
5. Distributed databases (MongoDB, Cassandra)
6. Internet services (Google, Facebook, Netflix)
7. Blockchain systems



PREREQUISITE COURSES

- Prerequisite course for Crypto Currency will be:
  1. Operating Systems: Understanding how processes, threads, memory, and system resources are managed, which is essential for coordinating programs running across multiple machines.
  2. Computer Networks: Understanding how computers communicate using network protocols such as TCP/IP and HTTP, which enables interaction between distributed system components.
  3. Data Structures and Algorithms: Understanding how data is efficiently organized, stored, and processed, which supports scalability and performance in distributed environments.
  4. Programming Skills: Ability to develop software using programming languages such as Java, C#, or Python, which is necessary to implement distributed services and communication logic.
  5. Database Fundamentals: Understanding how data is stored, retrieved, and managed, which is essential for handling distributed data, replication, and consistency.

TEXTBOOKS AND RELATED REFERENCES



COMMUNICATION PLATFORM

- Official Communication Platform will be Through Email by Group Leader

GRADING POLICIES

- Grading Policies will be according to the following:
  1. 50% For Final Exam
  2. 30% For Midterm
  3. 20% Presentations and Tasks.

SUMMARY

1. Mutual Introduction
2. Lecture Objectives
3. Introduction to Distributed Systems
4. Course Objectives
5. Course Outcomes
6. Real-Life Applications
7. Prerequisite Courses
8. Textbooks and References
9. Additional References
10. Communication Platform
11. Grading Policies